

Comparative Analysis of Airbnb Market Trends: Bangalore vs Mumbai

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Title : Comparative Analysis of Airbnb Market Trends: Bangalore vs Mumbai

Project Domain : Real Estate and Hospitality

Dataset Timeline : Y2025

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Mentor Name : Kumaran M

Raw Dataset :
[https://docs.google.com/spreadsheets/d/1YhWa5YlMmjL-rxm8xh7Qa9MVLx71NZ10/edit?usp=drive link&oid=105421993733585237367&rtpof=true&sd=true](https://docs.google.com/spreadsheets/d/1YhWa5YlMmjL-rxm8xh7Qa9MVLx71NZ10/edit?usp=drive_link&oid=105421993733585237367&rtpof=true&sd=true)

Cleaned Dataset :
[https://drive.google.com/file/d/1pQNsFWR3ATHo4HRHM22lbyXaD69_0Jeh/view?usp=drive link](https://drive.google.com/file/d/1pQNsFWR3ATHo4HRHM22lbyXaD69_0Jeh/view?usp=drive_link)

Introduction

This project belongs to the domain of Data Analytics with applications in Real Estate and Hospitality analytics, focusing on market pricing behaviour and demand patterns

Objective

- To perform an exploratory data analysis (EDA) and comparative study of Airbnb listings in Bangalore and Mumbai
- To uncover insights into pricing patterns, availability, customer preferences, and host behaviour
- To identify key factors influencing listing prices and occupancy
- To compare market dynamics between the two cities

Outcome

The project helps to identify trends, demand patterns, host behaviour and hidden insights from the dataset. The analysis derived actionable insights on factors affecting price and availability to support data-driven decision-making.

Dataset Information

Source: Doorstep Analytics

Year / Timeline: Data collected on 2025

Dataset Description

- Location: The primary metropolitan market (Bengaluru or Mumbai), used as the main categorical variable for comparison.
- City: A sub-region or secondary division within city.
- Neighbourhood: The specific local area.
- Lat: The exact Latitude coordinate used for mapping and identifying property density.
- Lng: The exact Longitude coordinate used for mapping and identifying property density.
- BasicNightPrice: The standard nightly rate in INR.
- Taxes: Additional government or service fees applied to the nightly rate
- Total_Price: A calculated feature ($\text{BasicNightPrice} + \text{Taxes}$) representing the true cost to the guest and revenue for the host
- RoomType: Categorization of the rental space (Entire Home, Private Room, or Shared Room)

- Space Type: Specific description of the property architecture (e.g., Apartment, Villa, or Boutique Hotel).
- Bathrooms: The total number of bathrooms available, used to determine the luxury level of the listing
- Bedrooms: The total number of dedicated bedrooms, a primary driver of the nightly price.
- Beds: The count of actual sleeping arrangements provided in the listing,
- Host is Super host: A binary indicator of professional host status, used to measure the "Super host Premium" in pricing

Type of Analysis

The analysis shows how Airbnb listings are distributed across Bangalore and Mumbai, highlights average pricing levels, and identifies the most common room types. It also provides insights into availability patterns and listing density across different neighbourhoods

The diagnostic analysis explores the reasons behind observed patterns in the data. It examines how different factors influence pricing and availability. The analysis reveals that prices vary significantly based on location, room type, and property features.

Although this project does not implement a predictive model, the dataset can be used to estimate future pricing trends. By using features such as location, room type, and number of guests, it is possible to build a model to predict Airbnb prices or identify listings with higher demand. This can help hosts make better pricing decisions.

Based on the insights derived from the analysis, several recommendations can be made. Hosts can optimize pricing strategies by adjusting rates according to location and property type. Listings with high availability but low reviews may require better pricing or improved services to attract customers. Additionally, focusing on popular neighbourhoods and competitive pricing can help increase bookings and revenue.

Stages for DA Project

Stage 1 – Problem Definition and Dataset Selection

The objective of this project is to perform a comparative analysis of Airbnb listings across two major cities to understand differences in pricing behaviour, availability, and customer preferences.

The dataset consists of Airbnb listing data from two metropolitan cities including features such as price, location, room type, availability, and review metrics.

Choose dataset and explain its source, size, and features Import libraries, load dataset. Dataset description (rows, columns, features) Initial EDA (head, info, describe, shape, null checks, duplicate check) import pandas

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
#Load the datasets
bang=pd.read_csv('https://raw.githubusercontent.com/reejariyaz/Comparative-Analysis-of-Airbnb-Market-Trends-Bangalore-Mumbai/refs/heads/main/DoorstepAnalytics_Airbnb_Bengaluru_India_Overview.csv')
mum=pd.read_csv('https://raw.githubusercontent.com/reejariyaz/Comparative-Analysis-of-Airbnb-Market-Trends-Bangalore-Mumbai/refs/heads/main/DoorstepAnalytics_Airbnb_Mumbai_India_Overview.csv')

#Append the two datasets together
df=pd.concat([bang,mum],ignore_index=True)
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 13226 entries, 0 to 13225

Data columns (total 44 columns):

#	Column	Non-Null Count	Dtype
0	Country	13226 non-null	object
1	Location	13226 non-null	object
2	Airbnb_ListingID	13226 non-null	int64
3	Scrape_Date	13226 non-null	object

4	Lat	13226 non-null float64
5	Lng	13226 non-null float64
6	City	13221 non-null object
7	LocalizedCity	13226 non-null object
8	LocalizedNeighbourhood_Source	9793 non-null object
9	LocalizedNeighbourhood_ML	9117 non-null object
10	LocalizedNeighbourhood	9981 non-null object
11	ListingTitle	13226 non-null object
12	Bathrooms	13186 non-null float64
13	Bedrooms	13174 non-null float64
14	Beds	13191 non-null float64
15	MaxGuests	13226 non-null int64
16	Host_ID	13226 non-null int64
17	Host_FirstName	13206 non-null object
18	Host_isSuperhost	13226 non-null bool
19	isNewListing	13206 non-null object
20	isSuperhost	13226 non-null bool
21	RoomType	13226 non-null object
22	RoomType_Clean	13184 non-null object
23	SpaceType	13226 non-null object
24	SpaceType_Clean	13226 non-null object
25	isHomeOrServicedApartment	13226 non-null int64
26	PictureCount	13226 non-null int64
27	ReviewCount	13226 non-null int64
28	ReviewScore	7437 non-null float64
29	StarRating	7447 non-null float64
30	isLimitedAirbnbData	13226 non-null bool
31	BasicNightPrice_CheckIn	12711 non-null object
32	BasicNightPrice_CheckOut	12711 non-null object
33	Currency	13226 non-null object

```
34 BasicNightPrice          12701 non-null float64
35 Taxes                    12541 non-null float64
36 BasicNightPriceString_USD 12701 non-null object
37 BasicNightPrice_USD      12701 non-null float64
38 Taxes_USD                12541 non-null float64
39 isOverpricedPerBeds_Calc  13226 non-null int64
40 isOverpricedPerGuest_Calc 13226 non-null int64
41 Amenities_SelfCheckIn    13194 non-null object
42 HostLanguages_Verbose    13226 non-null object
43 RecordInserted           13226 non-null object
```

dtypes: bool(3), float64(11), int64(8), object(22)

memory usage: 4.2+ MB

```
df.head()
```

	Country	Location	Airbnb_ListingID	Scrape_Date	Lat	Lng	City	LocalizedCity	LocalizedNeighbourhood_Source
0	India	Bengaluru	13183709	2025-10-31	13.021659	77.628189	Bengaluru	Bengaluru	Lingarajapura
1	India	Bengaluru	654064165740107808	2025-10-31	13.015500	77.671480	Bengaluru	Bengaluru	Ramamurthy Nagar
2	India	Bengaluru	1411530165672374120	2025-10-31	12.867241	77.557969	Bengaluru	Bengaluru	NaN
3	India	Bengaluru	920610774197649433	2025-10-31	12.982720	77.704970	Bengaluru	Bengaluru	Mahadevapura
4	India	Bengaluru	716367382981245820	2025-10-31	12.983480	77.705840	Bengaluru	Bengaluru	Mahadevapura

5 rows × 44 columns

```
df.tail()
```

	Country	Location	Airbnb_ListingID	Scrape_Date	Lat	Lng	City	LocalizedCity	LocalizedNeighbourhood_Source
13221	India	Mumbai	1389581808948528015	2025-10-29	19.128860	72.880440	Mumbai	Mumbai	Andheri East
13222	India	Mumbai	1331796873445309167	2025-10-29	19.131123	72.824684	Mumbai	Mumbai	Andheri West
13223	India	Mumbai	53055813	2025-10-29	19.079660	72.849550	Mumbai	Mumbai	Santacruz East
13224	India	Mumbai	1094726140998781775	2025-10-29	19.004801	72.829827	Mumbai	Mumbai	Lower Parel
13225	India	Mumbai	1337596129421155238	2025-10-29	19.112336	72.838016	Mumbai	Mumbai	Ville Parle West

5 rows × 44 columns

df.describe()

	Airbnb_ListingID	Lat	Lng	Bathrooms	Bedrooms	Beds	MaxGuests	Host_ID
count	1.322600e+04	13226.000000	13226.000000	13186.000000	13174.000000	13191.000000	13226.000000	1.322600e+04
mean	1.018637e+18	14.900869	76.120689	1.639087	1.653029	2.018573	3.802737	3.473174e+08
std	5.188617e+17	2.854215	2.222771	1.755579	2.022064	2.716814	2.787229	2.215514e+08
min	1.332000e+04	12.800650	72.783400	0.000000	0.000000	0.000000	1.000000	1.965400e+04
25%	8.048692e+17	12.921627	72.873323	1.000000	1.000000	1.000000	2.000000	1.507255e+08
50%	1.211341e+18	13.017075	77.602865	1.000000	1.000000	1.000000	3.000000	3.446852e+08
75%	1.424936e+18	19.064059	77.643644	2.000000	2.000000	2.000000	4.000000	5.444187e+08
max	1.542773e+18	19.210570	77.753628	50.000000	50.000000	81.000000	16.000000	7.260963e+08

df.shape

(13226, 44)

df.columns

```
Index(['Country', 'Location', 'Airbnb_ListingID', 'Scrape_Date', 'Lat', 'Lng',
      'City', 'LocalizedCity', 'LocalizedNeighbourhood_Source',
      'LocalizedNeighbourhood_ML', 'LocalizedNeighbourhood', 'ListingTitle',
      'Bathrooms', 'Bedrooms', 'Beds', 'MaxGuests', 'Host_ID',
      'Host_FirstName', 'Host_isSuperhost', 'isNewListing', 'isSuperhost',
      'RoomType', 'RoomType_Clean', 'SpaceType', 'SpaceType_Clean',
      'isHomeOrServicedApartment', 'PictureCount', 'ReviewCount',
      'ReviewScore', 'StarRating', 'isLimitedAirbnbData',
      'BasicNightPrice_CheckIn', 'BasicNightPrice_CheckOut', 'Currency',
      'BasicNightPrice', 'Taxes', 'BasicNightPriceString_USD',
      'BasicNightPrice_USD', 'Taxes_USD', 'isOverpricedPerBeds_Calc',
      'isOverpricedPerGuest_Calc', 'Amenities_SelfCheckIn',
      'HostLanguages_Verbose', 'RecordInserted'],
      dtype='object')
```

df.isnull().sum()

	0
Country	0
Location	0
Airbnb_ListingID	0
Scrape_Date	0
Lat	0
Lng	0
City	5
LocalizedCity	0

df.duplicated().sum()

np.int64(0)

Stage 2 – Data Cleaning and Pre-processing

The dataset initially contained multiple redundant and irrelevant columns, including identifiers, duplicate location fields, and columns with a high percentage of missing values. These were removed to improve data quality and focus the analysis on key variables such as pricing, location, and listing characteristics.

The city column had different formats and languages like Hindi. So, I converted all values into standard names like Bengaluru and Mumbai to make the data consistent.

Standardize all variations of city

```
diff=df[df['City']!=df['LocalizedCity']]
print(diff[['City','LocalizedCity']])
df.loc[(df['City'].str.contains('Bang|Beng',case=False,na=False))|
(df['LocalizedCity'].str.contains('Bang|Beng|Kar',case=False,na=False)), 'City']='Bengaluru'
df.loc[(df['City'].str.contains('Mumb|Mahar|मुंबई',case=False,na=False))|
(df['LocalizedCity'].str.contains('Mumb|Mahar|मुंबई',case=False,na=False)), 'City']='Mumbai'

diff=df[df['City']!=df['LocalizedCity']]
print(diff[['City','LocalizedCity']])
```

Interpretation

- This step addresses inconsistencies in the geographical data by standardizing various naming conversions for cities.

Removed unnecessary columns

```
df=df.drop(['Country','Airbnb_ListingID','Scrape_Date','LocalizedCity','LocalizedNeighbourhood_Source','LocalizedNeighbourhood_ML',
'ListingTitle','MaxGuests','Host_FirstName','Host_isSuperhost','RoomType_Clean','SpaceType_Clean','PictureCount',
'ReviewCount','ReviewScore','StarRating','isLimitedAirbnbData','BasicNightPrice_CheckIn','BasicNightPrice_CheckOut',
'Currency','BasicNightPriceString_USD','BasicNightPrice_USD','Taxes_USD','isOverpricedPerBeds_Calc','isOverpricedPerGuest',
'Amenities_SelfCheckIn','HostLanguages_Verbose','RecordInserted'],axis=1)
df.info()
```

This step involves the strategic removal of columns that do not contribute to the analytical goals of the project. By filtering out, the dataset becomes more focused and manageable.

Handling Duplicates

```
df.duplicated().sum()

np.int64(125)

df=df.drop_duplicates()

df.duplicated().sum()

np.int64(0)
```

Removing these duplicates ensures that datasets are cleaned, accurate and non-repetitive data.

Handling Missing Values

```
df['LocalizedNeighbourhood']=df['LocalizedNeighbourhood'].fillna('Unknown')
df.loc[
    (df['Location']=='Bengaluru') & (df['LocalizedNeighbourhood']=='Unknown'),
    'LocalizedNeighbourhood']='Unknown-Bengaluru'
df.loc[
    (df['Location']=='Mumbai') & (df['LocalizedNeighbourhood']=='Unknown'),
    'LocalizedNeighbourhood']='Unknown-Mumbai'
```

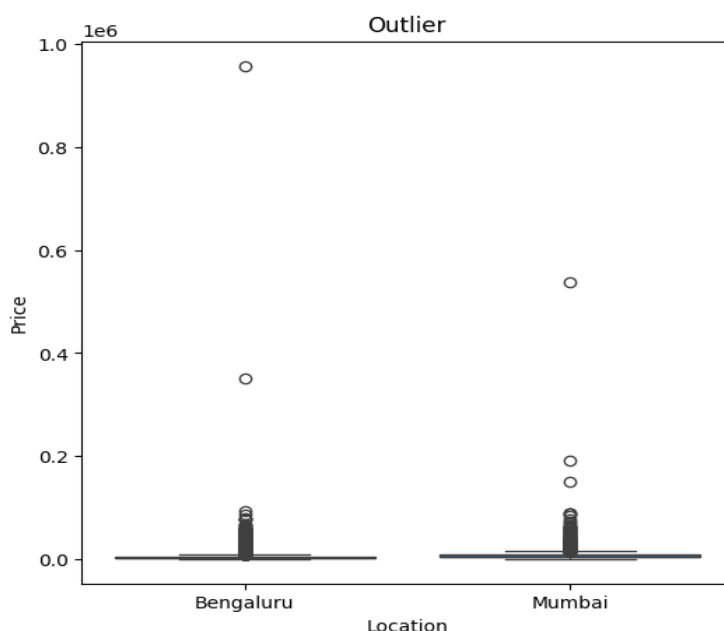
Missing values in the neighbourhood column were standardized as 'Unknown' and further categorized into city specific labels to preserve geographic integrity and improve comparative analysis between cities.

```
df['Bathrooms']=df['Bathrooms'].fillna(df['Bathrooms'].median())
df['Bedrooms']=df['Bedrooms'].fillna(df['Bedrooms'].median())
df['Beds']=df['Beds'].fillna(df['Beds'].median())
```

This step addresses missing values in key listing characteristics of bathrooms, bedrooms, and beds by imputing them with the median value of each respective column.

Treating Outliers

```
plt.figure(figsize=(6,6))
sns.boxplot(x='Location',y='BasicNightPrice',data=df)
plt.title('Outlier')
plt.ylabel('Price')
plt.show()
```



Treating outlier using IQR method

```
cleaned_data=[]
for Location in df['Location'].unique():
    city_data=df[df['Location']==Location]
    Q1=city_data['BasicNightPrice'].quantile(0.25)
    Q3=city_data['BasicNightPrice'].quantile(0.75)
    IQR=Q3-Q1
    lower_bound=Q1-1.5*IQR
    upper_bound=Q3+1.5*IQR

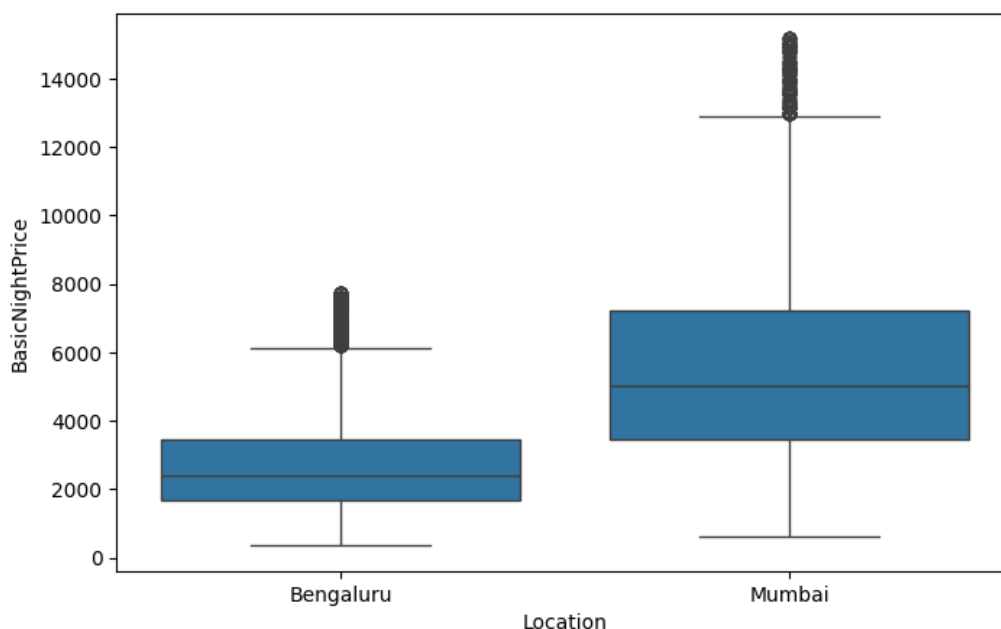
    print(f'Q1:{Q1},Q3:{Q3},IQR:{IQR}')
    print(f'The cut off for Outliers is:{upper_bound}')

    city_clean=city_data[(city_data['BasicNightPrice']>= lower_bound) &
                          (city_data['BasicNightPrice']<=upper_bound)]
    print(f'Original Rows:{len(city_data)}')
    print(f'Rows after IQR cleaning:{len(city_clean)}')

    cleaned_data.append(city_clean)
clean_df=pd.concat(cleaned_data)
```

After Treating Outlier-Box Plot

```
plt.figure(figsize=(8,5))
sns.boxplot(data=clean_df,x='Location',y='BasicNightPrice')
```



Outliers were removed using the IQR method applied separately for each city. The cleaned dataset was then used to visualize the distribution for accurate comparison of pricing between Bengaluru and Mumbai.

Fill with median

```
#Bengaluru Median
blr_median=df[df['Location']=='Bengaluru']['BasicNightPrice'].median()
#Mumbai Median
mum_median=df[df['Location']=='Mumbai']['BasicNightPrice'].median()

print(f'Remaining Nulls in BasicNightPrice of Bengaluru:{df[(df['Location']=='Bengaluru')]['BasicNightPrice'].isnull().sum()}')
df.loc[(df['Location']=='Bengaluru')&(df['BasicNightPrice'].isna()),'BasicNightPrice']=blr_median
print(f'Every missing price has been filled with the Bengaluru median:{blr_median}')

print(f'Remaining Nulls in BasicNightPrice of Mumbai:{df[(df['Location']=='Mumbai')]['BasicNightPrice'].isnull().sum()}')
df.loc[(df['Location']=='Mumbai')&(df['BasicNightPrice'].isna()),'BasicNightPrice']=mum_median
print(f'Every missing price has been filled with the Bengaluru median:{mum_median}')
```

```
Remaining Nulls in BasicNightPrice of Bengaluru:341
Every missing price has been filled with the Bengaluru median:2481.78
Remaining Nulls in BasicNightPrice of Mumbai:177
Every missing price has been filled with the Bengaluru median:5337.549999999999
```

Missing values in the BasicNightPrice column were imputed using the median value calculated at the city level.

Adding new column

```
clean_df['Total_Price']=clean_df['BasicNightPrice']+clean_df['Taxes']
```

A new feature, Total_Price was created by combining the BasicNightPrice and applicable taxes to represent the overall cost of a stay.

Now the datasets are cleaned successfully and ready for the analysis.

Uploaded the Dataset

```
clean_df.to_csv('cleaned_dataset.csv')
```

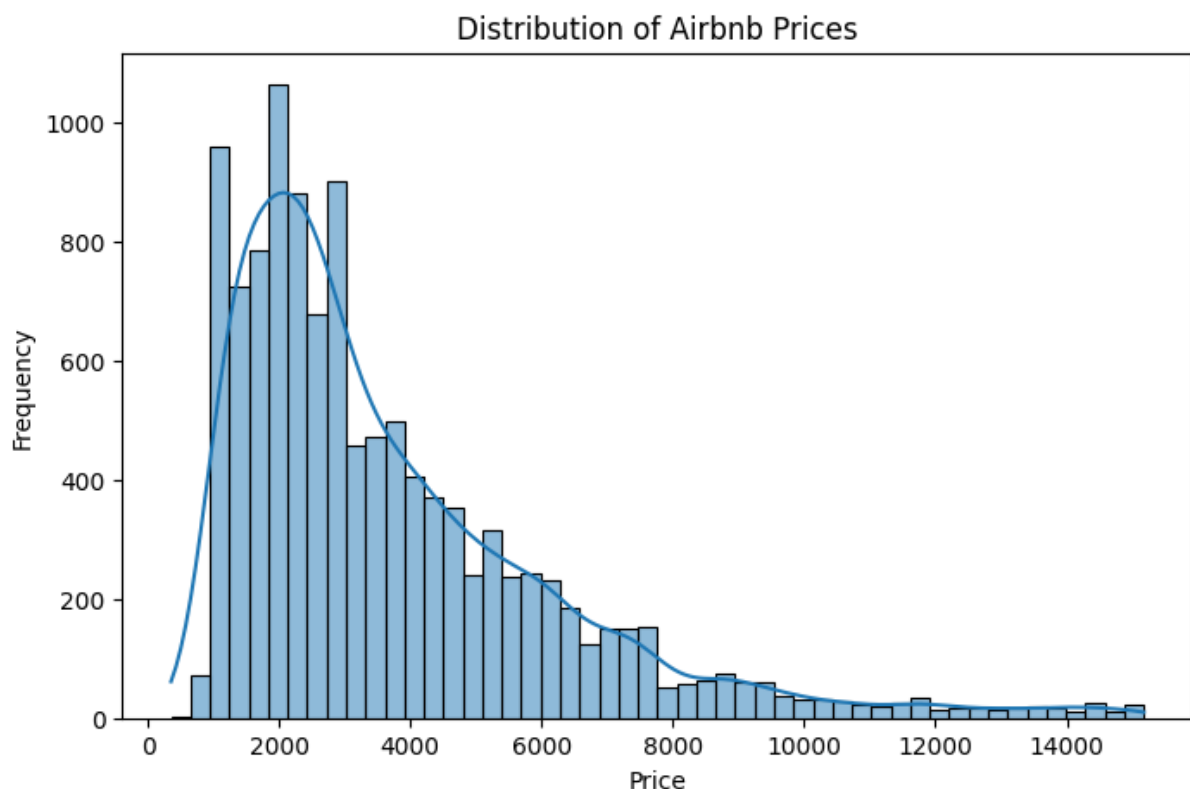
Exporting the cleaned Airbnb dataset for Bengaluru and Mumbai to a CSV file.
By using cleaned dataset ,created dashboard as follows.

Stage 3 – EDA and Visualizations

Univariate Analysis

Price Distribution

```
plt.figure(figsize=(8,5))
sns.histplot(clean_df['BasicNightPrice'],bins=50,kde=True)
plt.title('Distribution of Airbnb Prices')
plt.xlabel('Price')
plt.ylabel('Frequency')
plt.show()
```



The Histogram Chart shows the price distribution of Airbnb listings

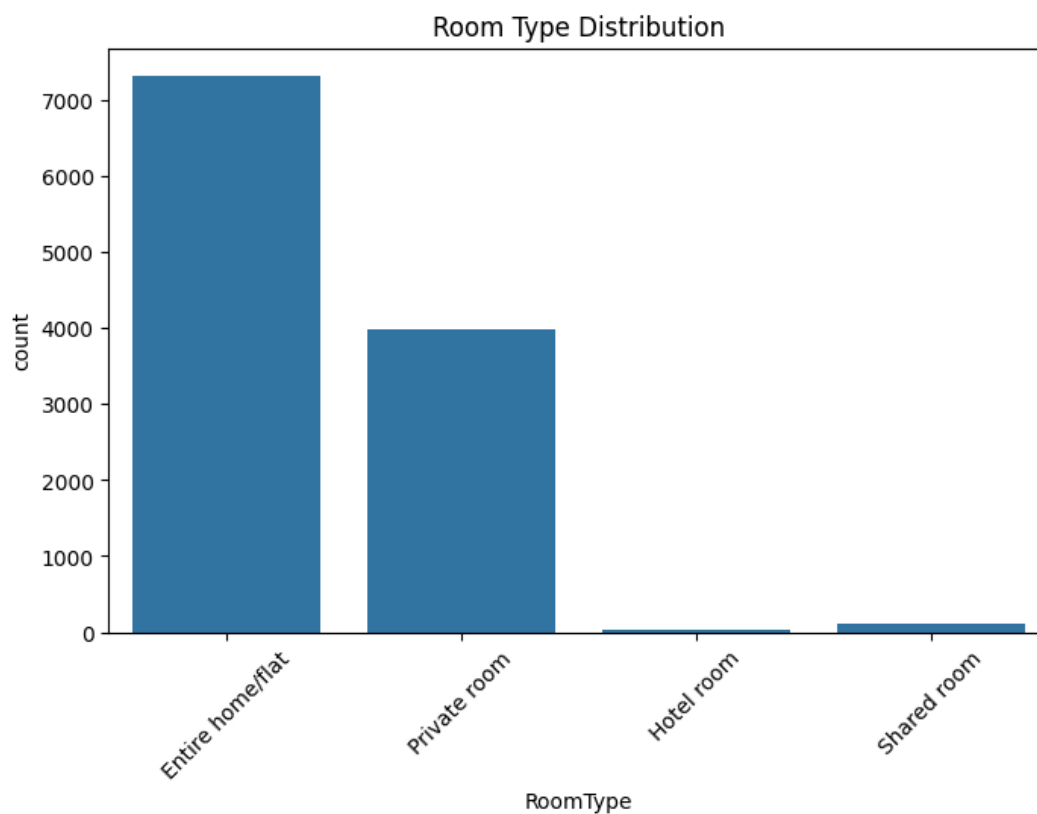
Feature used: BasicNightPrice represents the cost of a basic night's stay

Interpretations:

- The Chart is positive skewed; the highest peak occurs between 1500 and 2000.
- Most listings are priced under 6000, After that frequency drops significantly, showing that high-end rentals are much rarer.

Room Type Distribution

```
plt.figure(figsize=(8,5))
sns.countplot(x=clean_df['RoomType'])
plt.title('Room Type Distribution')
plt.xticks(rotation=45)
plt.show()
```



The CountPlot Chart is used to visualize the frequency of Airbnb listings belongs to each specific room category.

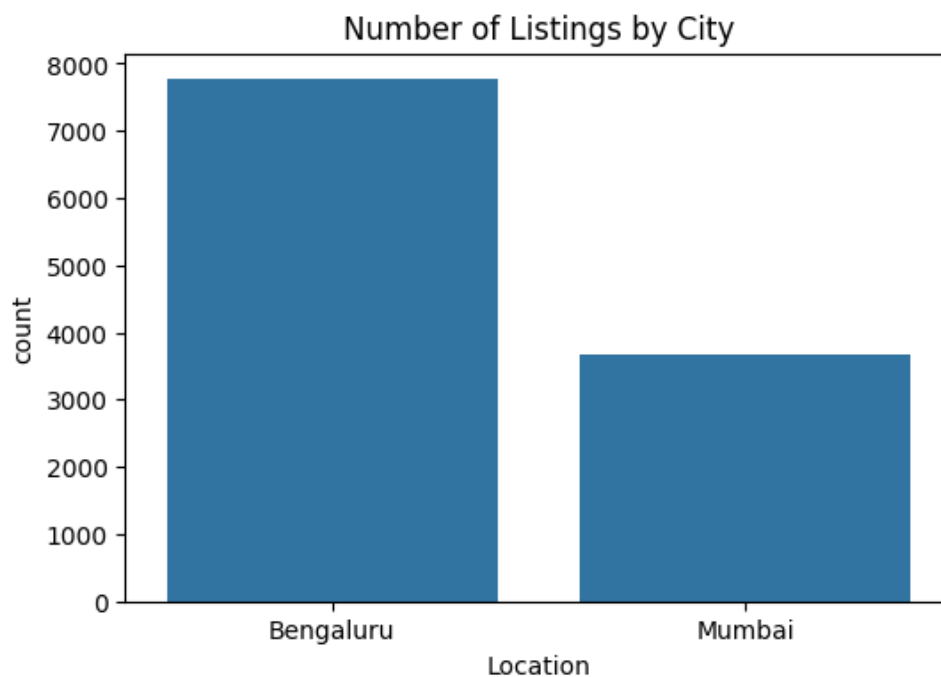
Feature used: RoomType represents the four categories: Entire home/flat, Private room, Hotel room, Shared room.

Interpretations:

The Chart reveals that Entire home/flat is the most preferred category, followed by Private room.

City-wise Distribution

```
plt.figure(figsize=(6,4))
sns.countplot(x=clean_df['Location'])
plt.title('Number of Listings by City')
plt.show()
```



The Bar Chart is used to visualize the distribution of Airbnb listings locations to identify which city has the highest market activity.

Feature used: Location represents the Bengaluru and Mumbai cities.

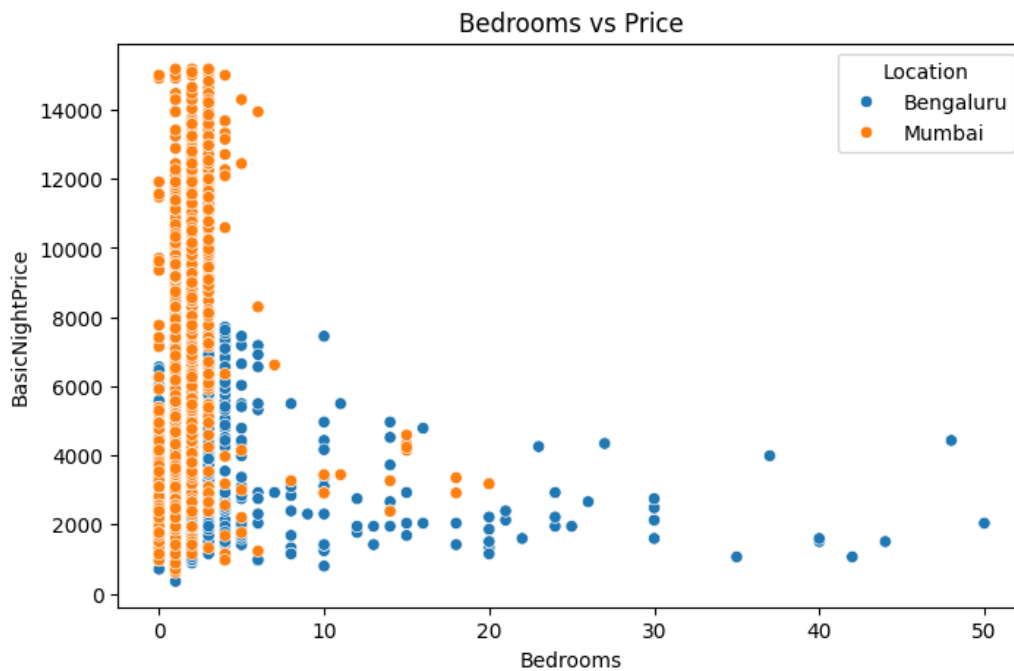
Interpretations:

Bengaluru has a significantly higher volume of listings, nearly 8000, which is nearly two times the amount found in Mumbai. It indicates much more active market in Bengaluru compared to Mumbai.

Bivariate Analysis

Relationship between Bedrooms and Price

```
plt.figure(figsize=(8,5))
sns.scatterplot(x='Bedrooms',y='BasicNightPrice',hue='Location',data=clean_df)
plt.title("Bedrooms vs Price")
plt.show()
```



The Scatter Plot Chart is used to observe the distribution of property prices relative to the number of bedrooms.

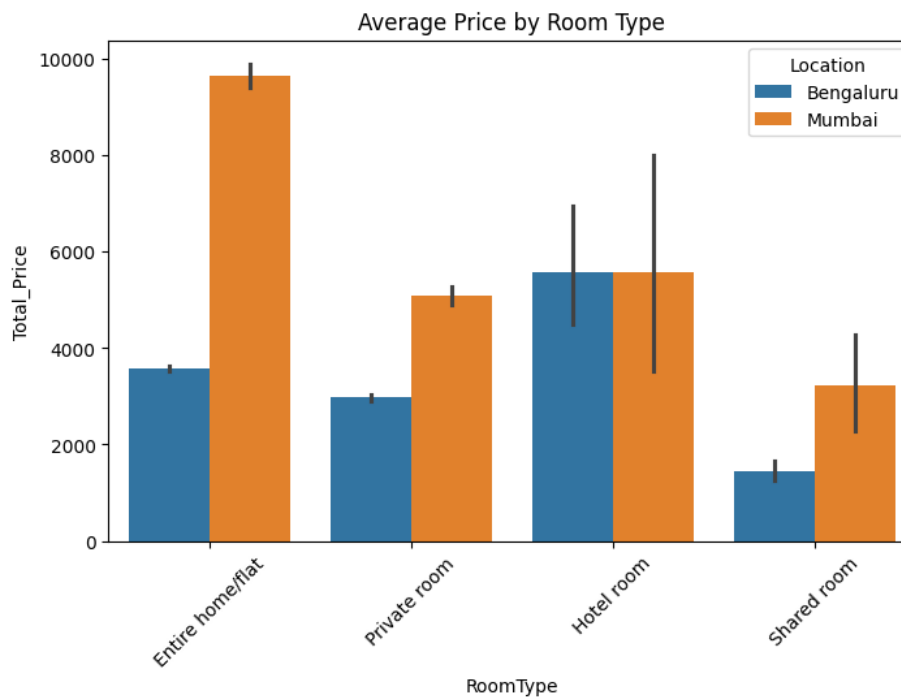
Feature used: Bedrooms represents the count of rooms. BasicNightPrice represents the price of bedroom.

Interpretations:

The Chart reveals that Mumbai has a very high density of listings with a low bedroom count but a wide price range. In contrast, Bengaluru shows more diversity in property size, though these larger properties maintain a lower price range compared to the smaller, high-end listings in Mumbai.

Average price of both cities

```
plt.figure(figsize=(8,5))
sns.barplot(x='RoomType',y='Total_Price',hue='Location',data=clean_df.dropna(subset=['Total_Price']))
plt.title('Average Price by Room Type')
plt.xticks(rotation=45)
plt.show()
```



The Bar Plot Chart purpose is to compare the average of Total_Price across different RoomType simultaneously comparing how these values differ between two specific Location.

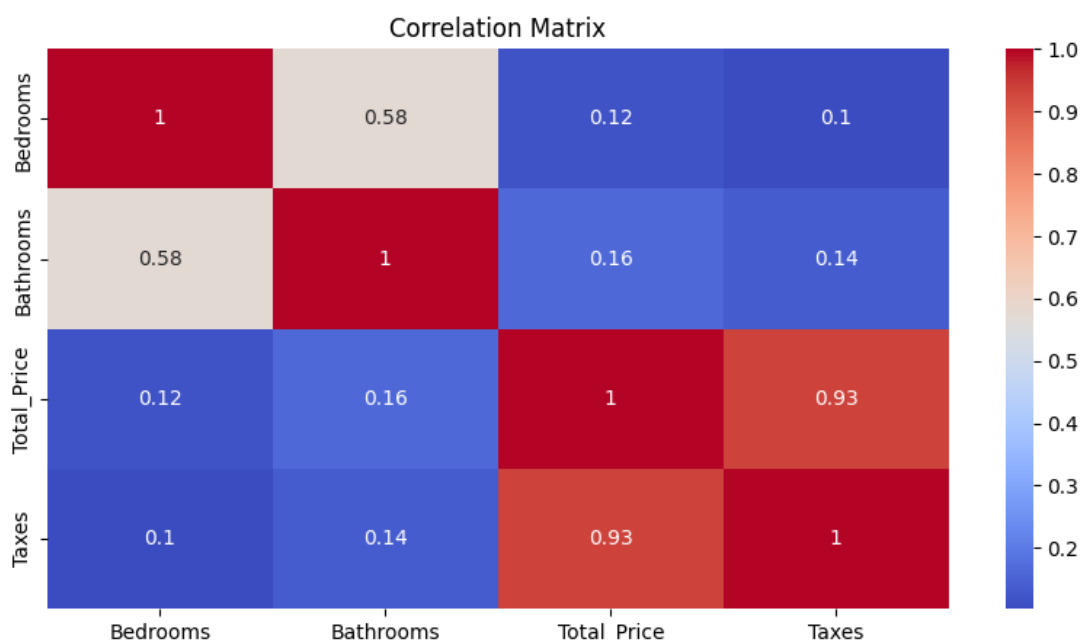
Feature used: RoomType represents the four categories: Entire home/flat, Private room, Hotel room, Shared room. Total_Price represents the average calculated cost.

Interpretations:

Mumbai consistently has higher average total prices than Bengaluru for almost all room categories, most notably for Entire home/flat where the price is more than double. Hotel rooms show the most significant price variability in both cities, suggesting that hotel pricing is much less standardized than private or shared room listings.

Correlation Heatmap

```
plt.figure(figsize=(10,5))
corr=clean_df.dropna(subset=['Total_Price','Taxes'])[['Bedrooms','Bathrooms','Total_Price','Taxes']].corr()
sns.heatmap(corr,annot=True,cmap='coolwarm')
plt.title('Correlation Matrix')
#plt.xticks(rotation=45)
plt.show()
```



The Correlation Heatmap is used to visualize how strongly different numerical features are related to each other after data cleaning.

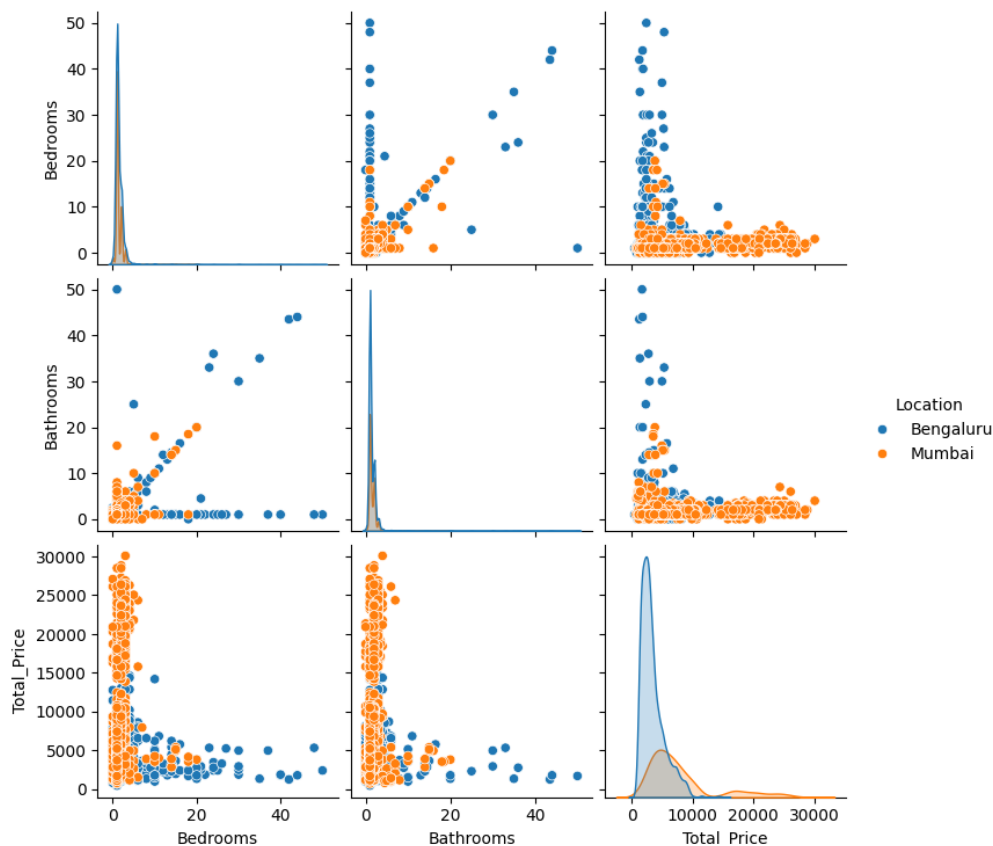
Feature used: Bedrooms, Bathrooms, Total_Price, and Taxes.

Interpretations:

The Chart reveals a very strong positive correlation between Total Price and Taxes, meaning they increase directly together. However there is only a weak correlation between the number of rooms and the price.

Multivariate Analysis

```
plt.figure(figsize=(8,6))
plot_data=clean_df.dropna(subset=['Total_Price'])
sns.pairplot(plot_data[['Bedrooms', 'Bathrooms', 'Total_Price', 'Location']],hue='Location')
plt.show()
```



The Seaborn Pair Plot chart is used to visualize the pairwise relationships between multiple numeric variables. It color-codes the data by Location to show how these cities differ across all features at once.

Feature used: Bedrooms, Bathrooms, Total_Price and Location for hue.

Interpretations:

- Bengaluru generally has a higher frequency of lower priced listings compared to Mumbai.
- There is a strong positive correlation between the number of bedrooms and bathrooms in both cities.
- Surprisingly, a higher number of bedrooms or bathrooms does not automatically mean a higher price. Many listings with many bedrooms in Bengaluru are priced lower than smaller luxury listings.
- Bengaluru listings appear to have a wider range of property sizes, whereas Mumbai listings are more tightly clustered in terms of size but contain competitive pricing.

Grouped Analysis-Advanced Plot

```
pivot=clean_df.pivot_table(values='Total_Price',index='Location',columns='RoomType',aggfunc='mean')  
print(pivot)
```

RoomType	Entire home/flat	Hotel room	Private room	Shared room
Location				
Bengaluru	3567.138531	5562.735600	2978.161180	1448.300200
Mumbai	9648.277097	5563.992667	5097.807711	3218.705333

```
pivot.plot(kind='bar',figsize=(8,5))  
plt.title("City vs Room Type Price Comparison")  
plt.ylabel('Average Price')  
plt.show()
```



The Bar Chart visualizes the average price differences between Mumbai and Bengaluru across four specific room categories.

Feature used: Location and RoomType.

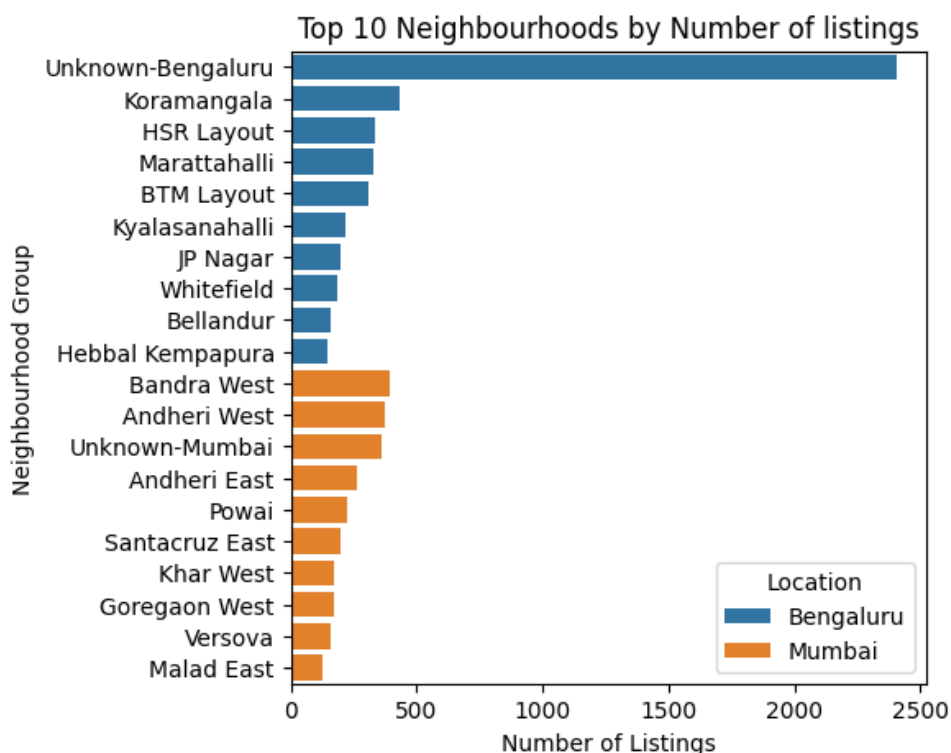
Interpretations:

- Mumbai is significantly more expensive than Bengaluru in every category, especially for Entire homes / flats, which cost nearly 3 times more.
- Bengaluru's most expensive category is Hotel rooms, not entire homes.
- A Shared room in Mumbai is nearly as expensive as an Entire home in Bengaluru.
- Shared rooms in Bengaluru represent the most affordable price point for travellers.

Neighbourhood Analysis

```
from pandas.core.groupby import groupby
plt.figure(figsize=(5,5))
top10_city=(clean_df.groupby(['Location','LocalizedNeighbourhood']).size()
              .reset_index(name='Count')
              .sort_values(['Location','Count'],ascending=[True,False])
)
top10_city=top10_city.groupby('Location').head(10)

sns.barplot(data=top10_city,x='Count',y='LocalizedNeighbourhood',hue='Location')
plt.title('Top 10 Neighbourhoods by Number of listings')
plt.xlabel('Number of Listings')
plt.ylabel('Neighbourhood Group')
plt.show()
```



The Horizontal Bar Chart identifies the Top 10 most popular neighbourhoods based on the Airbnb listings, categorized both cities.

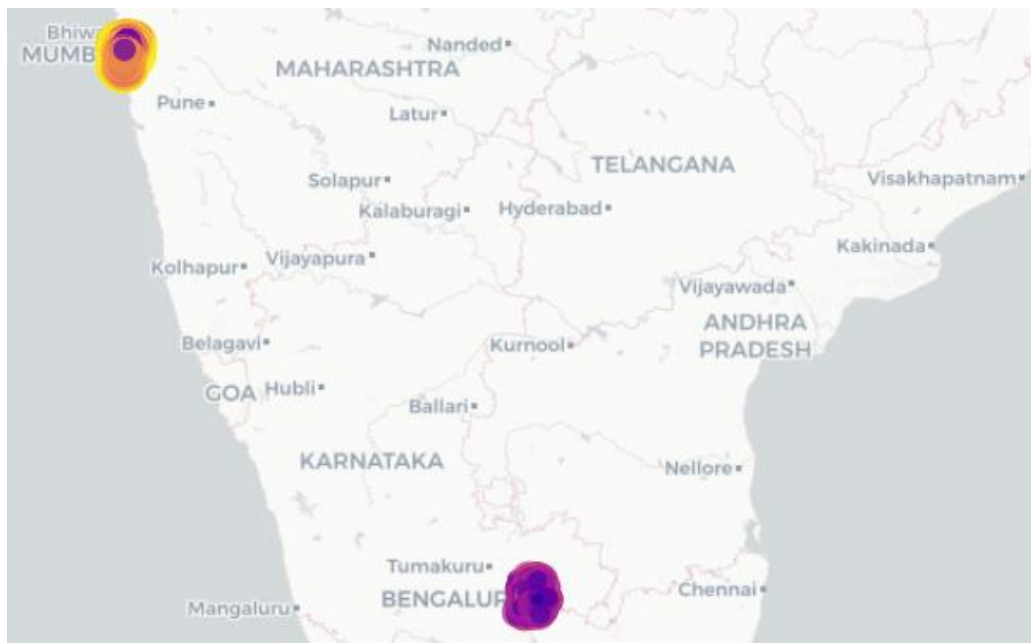
Feature used: Location and Localized Neighbourhood

Interpretations:

- Excluding Unknown, Koramangala is the leading neighbourhood with the highest number of listings, making it a primary Airbnb hotspot in Bengaluru.
- Popular areas like HSR Layout, Marathahalli, and BTM Layout indicating that areas concentrated in Tech and residential Hubs.
- In Mumbai, Andheri West and Bandra West are the most prominent neighbourhoods. These are well known commercial and entertainment hubs.

Geospatial Scatter Map

```
import plotly.express as px
fig=px.scatter_mapbox(
    clean_df,lat='Lat',lon='Lng',color='BasicNightPrice',size='BasicNightPrice',
    zoom=5,center={'lat':16.0,'lon':75.5},mapbox_style="carto-positron",size_max=20,title="Airbnb Listings Map"
)
fig.show()
```



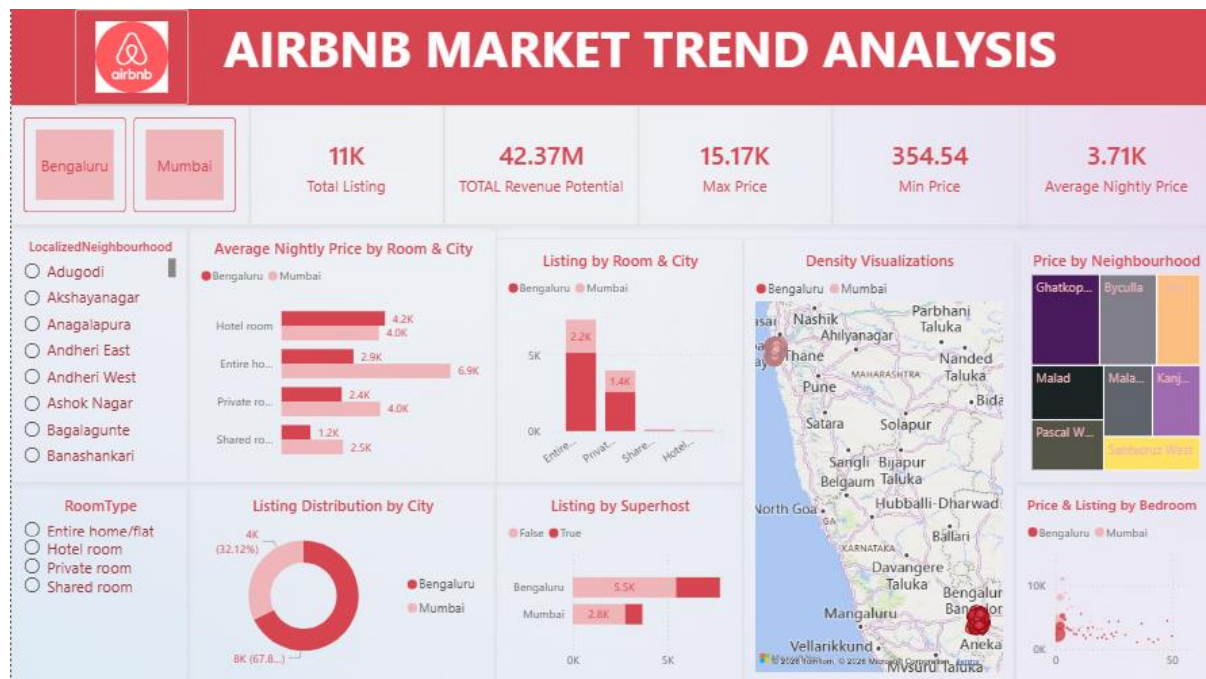
The Geospatial Scatter Map used to visualize the physical distribution of Airbnb listings across the geographical landscape of India.

Feature used: Latitude, Longitude, Location, BasicNightPrice.

Interpretations:

The colors intensity indicates that higher-priced listings are spread throughout both cities. Mumbai appears to have a very dense concentration of high value points.

Stage 4 : Dashboard



Summary

- The dataset contains roughly 11K listings across Bengaluru and Mumbai making it suitable for city-level Airbnb market analysis.
- The dashboard shows Total Revenue Potential = 42.37M.
- The Average Nightly Price = 3.71K, with a Maximum Price = 15.17K and Minimum Price = 354.54.
- Most listings are concentrated in Bengaluru, especially in areas like HSR Layout, Koramangala, BTM Layout, Whitefield, Bellandur, Domlur, Indira Nagar, and Marathahalli.
- The dataset includes key property attributes such as City, Neighbourhood, Bathrooms, Bedrooms, Beds, Room Type, Space Type, Superhost status, new listing status, BasicNightPrice, Taxes, and Total_Price.

Key Insights

- Entire homes/flats dominate the market, showing that guests and hosts prefer full-property stays over shared options.
- Private rooms and shared rooms play a smaller role in overall market revenue.
- Revenue and pricing vary strongly by neighbourhood, with premium localities charging much higher nightly rates than budget areas.
- Bengaluru has dense property concentration in central and tech-driven areas, showing strong demand around business and residential hubs.
- Listings with more bedrooms generally have higher total prices.
- Koramangala, HSR Layout, Whitefield, and Bellandur stand out as strategic zones because they combine strong location value with active listing density.
- Hotel-style rooms and serviced apartments can command higher rates in selected zones, especially near commercial areas.
- Premium neighbourhoods support premium pricing even for similar property types, proving that geography strongly influences demand.

Business Recommendations

- Focus on entire homes/flats in high-demand neighbourhoods. These listings should be the main growth target because they dominate demand and can generate stronger returns.
- Use location-based pricing. For example, properties in Koramangala, HSR Layout, Indira Nagar, Whitefield, and Bellandur can be priced above average.
- Hosts should improve furnishing, cleanliness, Wi-Fi, and photos to boost bookings.
- A small weekly or monthly discount can help improve occupancy in budget neighbourhoods.
- Improve listing presentation. Better photos, clear descriptions, and amenity highlights can raise conversion in dense markets where many properties look similar.
- Target business-travel zones. Areas near offices, tech parks, and transport access are more likely to sustain stable demand. Standardize cleaning and guest experience.

Conclusion

Airbnb success here depends mainly on location, property type, and pricing strategy. Properties in strong urban neighbourhoods, especially full homes with 1-3 bedrooms, have the best chance to earn more. In short, the winning formula is "Right place + Right property + Right price + Strong host performance". The Airbnb market is competitive but highly profitable when hosts focus on entire homes in high-demand neighbourhoods and price them according to local market value.